

Concept 5 | Inequalities

English Student Edition • Focused formulas and guided practice

Formula opener

Square rule

$$(x - y)^2 \geq 0$$

Order

$$0 < a < b \Rightarrow a^2 < b^2$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab}$$

Worked Solutions

Inequalities

Q001 Written

Under the condition $a > b$, $x > y$, prove that $2(ax + by) > (a + b)(x + y)$.

Solution

1. Start with LHS-RHS: $2(ax+by)-(a+b)(x+y)$.
2. Expand and factor: $ax-ay-bx+by=(a-b)(x-y)$.
3. Both factors are positive, so $(a-b)(x-y)>0$; hence the required inequality holds.

Final answer

$$2(ax + by) > (a + b)(x + y)$$

Q002 MCQ

If $x>y$, prove that $3x + y > 2(x + y)$.

If $x>y$, the positive difference between the two sides is:

Solution

1. Subtract the right-hand side: $x - y$.
2. Factor the difference as $x - y$.
3. The condition $x>y$ makes each factor positive.
4. Hence $3x + y > 2(x + y)$.

Correct option: A

Final answer

$$x - y$$

Worked Solutions

Inequalities

Q003 MCQ

If $x > y$, prove that $3x + 2y > 2x + 3y$.

If $x > y$, the positive difference between the two sides is:

Solution

1. Subtract the right-hand side: $x - y$.
2. Factor the difference as $x - y$.
3. The condition $x > y$ makes each factor positive.
4. Hence $3x + 2y > 2x + 3y$.

Correct option: C

Final answer

$$x - y$$

Q004 MCQ

If $x > y$, prove that $3(x + y) > 2(x + 2y)$.

If $x > y$, the positive difference between the two sides is:

Solution

1. Subtract the right-hand side: $x - y$.
2. Factor the difference as $x - y$.
3. The condition $x > y$ makes each factor positive.
4. Hence $3(x + y) > 2(x + 2y)$.

Correct option: D

Final answer

$$x - y$$

Worked Solutions

Inequalities

Q005 MCQ

If $x > y$, prove that $3x + 4y > 2x + 5y$.

If $x > y$, the positive difference between the two sides is:

Solution

1. Subtract the right-hand side: $x - y$.
2. Factor the difference as $x - y$.
3. The condition $x > y$ makes each factor positive.
4. Hence $3x + 4y > 2x + 5y$.

Correct option: D

Final answer

$$x - y$$

Q006 MCQ

If $x > y$, prove that $3x + 5y > 2(x + 3y)$.

If $x > y$, the positive difference between the two sides is:

Solution

1. Subtract the right-hand side: $x - y$.
2. Factor the difference as $x - y$.
3. The condition $x > y$ makes each factor positive.
4. Hence $3x + 5y > 2(x + 3y)$.

Correct option: B

Final answer

$$x - y$$

Worked Solutions

Inequalities

Q007 Written

Under the condition $a > b$, $c > d$, prove that $ac + bd > ad + bc$.

Solution

1. Subtract the right-hand side: $ac + bd - ad - bc$.
2. Factor: $a(c - d) - b(c - d) = (a - b)(c - d)$.
3. Both factors are positive, so the difference is positive; therefore $ac + bd > ad + bc$.

Final answer

$$ac + bd > ad + bc$$

Q008 Written

Under the condition $a > c$, $b > d$, prove that $ab + cd > ad + bc$.

Solution

1. Subtract the right-hand side: $ab + cd - ad - bc$.
2. Factor: $a(b - d) - c(b - d) = (a - c)(b - d)$.
3. Both factors are positive, so the difference is positive; hence $ab + cd > ad + bc$.

Final answer

$$ab + cd > ad + bc$$

Worked Solutions

Inequalities

Q009 MCQ

If $a > b$ and $c > d$, prove that $ac + bd > ad + bc$.

If $a > b$ and $c > d$, the positive difference between the two sides is:

Solution

1. Subtract the right-hand side: $(a - b)(c - d)$.
2. Factor the difference as $(a - b)(c - d)$.
3. The condition $a > b$ and $c > d$ makes each factor positive.
4. Hence $ac + bd > ad + bc$.

Correct option: D

Final answer

$$(a - b)(c - d)$$

Q010 MCQ

If $a > 2b$ and $c > d$, prove that $ac + 2bd > ad + 2bc$.

If $a > 2b$ and $c > d$, the positive difference between the two sides is:

Solution

1. Subtract the right-hand side: $(a - 2b)(c - d)$.
2. Factor the difference as $(a - 2b)(c - d)$.
3. The condition $a > 2b$ and $c > d$ makes each factor positive.
4. Hence $ac + 2bd > ad + 2bc$.

Correct option: C

Final answer

$$(a - 2b)(c - d)$$

Worked Solutions

Inequalities

Q011 MCQ

If $a > 3b$ and $c > d$, prove that $ac + 3bd > ad + 3bc$.

If $a > 3b$ and $c > d$, the positive difference between the two sides is:

Solution

1. Subtract the right-hand side: $(a - 3b)(c - d)$.
2. Factor the difference as $(a - 3b)(c - d)$.
3. The condition $a > 3b$ and $c > d$ makes each factor positive.
4. Hence $ac + 3bd > ad + 3bc$.

Correct option: A

Final answer

$$(a - 3b)(c - d)$$

Q012 Written

Under the condition $x > y$, prove that $3x - 4y > 2x - 3y$.

Solution

1. Subtract the right-hand side: $(3x - 4y) - (2x - 3y)$.
2. Simplify to $x - y$.
3. Since $x > y$, $x - y > 0$; therefore $3x - 4y > 2x - 3y$.

Final answer

$$3x - 4y > 2x - 3y$$

Worked Solutions

Inequalities

Q013 Written

Under the condition $x > 3$, $y > -5$, prove that $xy + 5x > 3y + 15$.

Solution

1. Start with LHS-RHS: $xy+5x-3y-15$.
2. Factor by grouping: $x(y+5) - 3(y+5) = (x-3)(y+5)$.
3. Since $x-3>0$ and $y+5>0$, the difference is positive; hence the inequality holds.

Final answer

$$xy + 5x > 3y + 15$$

Q014 Written

Under the condition $x > 2$, $y > 1$, prove that $xy + 2 > x + 2y$.

Solution

1. Start with LHS-RHS: $xy+2-x-2y$.
2. Factor: $x(y-1) - 2(y-1) = (x-2)(y-1)$.
3. Both factors are positive, so the difference is positive; hence $xy+2>x+2y$.

Final answer

$$xy + 2 > x + 2y$$

Worked Solutions

Inequalities

Q015 Written

Under the condition $a > 1$, $b > 1$, prove that $ab + 1 > a + b$.

Solution

1. Start with LHS-RHS: $ab+1-a-b$.
2. Factor: $ab-a-b+1=(a-1)(b-1)$.
3. Since $a-1>0$ and $b-1>0$, the difference is positive; hence $ab+1>a+b$.

Final answer

$$ab + 1 > a + b$$

Q016 Written

Under the condition $a > b > x > y$, prove that $ax + by > ay + bx$.

Solution

1. Subtract the right-hand side: $ax+by-ay-bx$.
2. Factor: $a(x - y) - b(x - y) = (a - b)(x - y)$.
3. Both factors are positive because $a>b$ and $x>y$; therefore the difference is positive.

Final answer

$$ax + by > ay + bx$$

Worked Solutions

Inequalities

Q017 **Written**

Under the condition $a > b > c > 0$, $x > y > z > 0$, prove that $ax + by + cz > ay + bz + cx$.

Solution

1. Subtract the right-hand side and regroup: $ax+by+cz-ay-bz-cx$.
2. Rewrite it as $(a-b)(x-y)+(b-c)(x-z)$.
3. Every factor in both products is positive, so the sum is positive; hence $ax+by+cz>ay+bz+cx$.

Final answer

$$ax + by + cz > ay + bz + cx$$

Worked Solutions

Inequalities

Q018 Challenge

If $a > b$ and $x > y$, prove that $2(ax + by) > 2(ay + bx)$.

If $a > b$ and $x > y$, the positive difference between the two sides is:

Solution

1. Subtract the right-hand side: $2(a - b)(x - y)$.
2. Factor the difference as $2(a - b)(x - y)$.
3. The condition $a > b$ and $x > y$ makes each factor positive.
4. Hence $2(ax + by) > 2(ay + bx)$.

Correct option: D

Final answer

$$2(a - b)(x - y)$$

Worked Solutions

Inequalities

Q019 Written

Prove that $a^2 + 4b^2 \geq 4ab$.**Solution**

1. Start with LHS - RHS: $a^2 + 4b^2 - 4ab$.
2. Complete the square: $a^2 + 4b^2 - 4ab = (a - 2b)^2 \geq 0$.
3. Therefore $a^2 + 4b^2 \geq 4ab$.
4. Equality holds when $a = 2b$.

Final answer

$$a^2 + 4b^2 \geq 4ab$$

Q020 MCQ

Under a,b real, prove that $a^2 + 4b^2 \geq 4ab$ **select the correct factorisation of LHS - RHS:****Solution**

1. Start with LHS - RHS: $a^2 + 4b^2 - 4ab$.
2. Factor/complete the square to obtain $(a - 2b)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $a^2 + 4b^2 \geq 4ab$.

Correct option: D**Final answer**

$$(a - 2b)^2$$

Worked Solutions

Inequalities

Q021 MCQ

Under a, b real, prove that $a^2 + 9b^2 \geq 6ab$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $a^2 + 9b^2 - 6ab$.
2. Factor/complete the square to obtain $(a - 3b)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $a^2 + 9b^2 \geq 6ab$.

Correct option: C**Final answer**

$$(a - 3b)^2$$

Q022 MCQ

Under a, b real, prove that $a^2 + 16b^2 \geq 8ab$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $a^2 + 16b^2 - 8ab$.
2. Factor/complete the square to obtain $(a - 4b)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $a^2 + 16b^2 \geq 8ab$.

Correct option: B**Final answer**

$$(a - 4b)^2$$

Worked Solutions

Inequalities

Q023 MCQ

Under a, b real, prove that $a^2 + 25b^2 \geq 10ab$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $a^2 + 25b^2 - 10ab$.
2. Factor/complete the square to obtain $(a - 5b)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $a^2 + 25b^2 \geq 10ab$.

Correct option: A

Final answer

$$(a - 5b)^2$$

Q024 MCQ

Under a, b real, prove that $a^2 + 36b^2 \geq 12ab$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $a^2 + 36b^2 - 12ab$.
2. Factor/complete the square to obtain $(a - 6b)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $a^2 + 36b^2 \geq 12ab$.

Correct option: C

Final answer

$$(a - 6b)^2$$

Worked Solutions

Inequalities

Q025 MCQ

Under a, b real, prove that $a^2 + 49b^2 \geq 14ab$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $a^2 + 49b^2 - 14ab$.
2. Factor/complete the square to obtain $(a - 7b)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $a^2 + 49b^2 \geq 14ab$.

Correct option: B**Final answer**

$$(a - 7b)^2$$

Q026 Written

Prove that $a^2 + 5b^2 \geq 4ab$.

Solution

1. Start with LHS - RHS: $a^2 + 5b^2 - 4ab$.
2. Complete the square: $a^2 + 5b^2 - 4ab = (a - 2b)^2 + b^2 \geq 0$.
3. Therefore $a^2 + 5b^2 \geq 4ab$.
4. Equality holds when $a = b = 0$.

Final answer

$$a^2 + 5b^2 \geq 4ab$$

Worked Solutions

Inequalities

Q027 **Written****Prove that** $9x^2 \geq y(6x - y)$.**Solution**

1. Subtract the right-hand side: $9x^2 - y(6x - y)$.
2. Complete the square: $9x^2 - y(6x - y) = (3x - y)^2 \geq 0$.
3. Therefore $9x^2 \geq y(6x - y)$.
4. Equality holds when $y = 3x$.

Final answer

$$9x^2 \geq y(6x - y)$$

Q028 **Written****Prove that** $2(x^2 + y^2) \geq (x + y)^2$.**Solution**

1. Subtract the right-hand side: $2(x^2 + y^2) - (x + y)^2$.
2. This is $(x - y)^2 \geq 0$.
3. Therefore the inequality holds.
4. Equality holds when $x = y$.

Final answer

$$2(x^2 + y^2) \geq (x + y)^2$$

Worked Solutions

Inequalities

Q029 Written

Prove that $(2x - y)^2 \geq 3(x^2 - y^2)$.**Solution**

1. Subtract the right-hand side: $(2x - y)^2 - 3(x^2 - y^2)$.
2. Simplify to $(x - 2y)^2 \geq 0$.
3. Therefore the inequality holds.
4. Equality holds when $x = 2y$.

Final answer

$$(2x - y)^2 \geq 3(x^2 - y^2)$$

Q030 Written

Prove that $(2a + 3b)^2 \geq 12ab$.**Solution**

1. Subtract the right-hand side: $(2a + 3b)^2 - 12ab$.
2. The cross terms cancel, giving $4a^2 + 9b^2 \geq 0$.
3. Therefore the inequality holds.
4. Equality holds when $a = b = 0$.

Final answer

$$(2a + 3b)^2 \geq 12ab$$

Worked Solutions

Inequalities

Q031 MCQ

Under x, y real, prove that $4x^2 \geq y(4x - y)$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $4x^2 - y(4x - y)$.
2. Factor/complete the square to obtain $(2x - y)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $4x^2 \geq y(4x - y)$.

Correct option: A**Final answer**

$$(2x - y)^2$$

Q032 MCQ

Under x, y real, prove that $9x^2 \geq y(6x - y)$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $9x^2 - y(6x - y)$.
2. Factor/complete the square to obtain $(3x - y)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $9x^2 \geq y(6x - y)$.

Correct option: C**Final answer**

$$(3x - y)^2$$

Worked Solutions

Inequalities

Q033 MCQ

Under x, y real, prove that $16x^2 \geq y(8x - y)$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $16x^2 - y(8x - y)$.
2. Factor/complete the square to obtain $(4x - y)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $16x^2 \geq y(8x - y)$.

Correct option: D**Final answer**

$$(4x - y)^2$$

Q034 MCQ

Under x, y real, prove that $25x^2 \geq y(10x - y)$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $25x^2 - y(10x - y)$.
2. Factor/complete the square to obtain $(5x - y)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $25x^2 \geq y(10x - y)$.

Correct option: D**Final answer**

$$(5x - y)^2$$

Worked Solutions

Inequalities

Q035 MCQ

Under x, y real, prove that $36x^2 \geq y(12x - y)$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $36x^2 - y(12x - y)$.
2. Factor/complete the square to obtain $(6x - y)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $36x^2 \geq y(12x - y)$.

Correct option: A

Final answer

$$(6x - y)^2$$

Q036 MCQ

Under x, y real, prove that $x^2 + y^2 \geq 2xy$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $x^2 + y^2 - 2xy$.
2. Factor/complete the square to obtain $(x - y)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $x^2 + y^2 \geq 2xy$.

Correct option: C

Final answer

$$(x - y)^2$$

Worked Solutions

Inequalities

Q037 Written

Prove that $x^2 + y^2 \geq 2(x - y - 1)$.**Solution**

1. Start with LHS - RHS: $x^2 + y^2 - 2(x - y - 1)$.
2. Complete the squares: $x^2 + y^2 - 2(x - y - 1) = (x - 1)^2 + (y + 1)^2 \geq 0$.
3. Therefore $x^2 + y^2 \geq 2(x - y - 1)$.
4. Equality holds when $x = 1$, $y = -1$.

Final answer

$$x^2 + y^2 \geq 2(x - y - 1)$$

Q038 Written

Prove that $x^2 + 2 > 2x$.**Solution**

1. Subtract the right-hand side: $x^2 + 2 - 2x$.
2. Complete the square: $x^2 + 2 - 2x = (x - 1)^2 + 1 > 0$.
3. Therefore $x^2 + 2 > 2x$.
4. There is no equality case.

Final answer

$$x^2 + 2 > 2x$$

Worked Solutions

Inequalities

Q039 Written

Prove that $x^2 + y^2 - 6x + 8y + 25 \geq 0$.**Solution**

1. Complete the squares: $x^2 + y^2 - 6x + 8y + 25 = (x - 3)^2 + (y + 4)^2 \geq 0$.
2. Therefore the inequality holds.
3. Equality holds when $x = 3$, $y = -4$.

Final answer

$$x^2 + y^2 - 6x + 8y + 25 \geq 0$$

Q040 Written

Prove that $x^2 + 2y^2 - 10x + 12y + 43 \geq 0$.**Solution**

1. Complete the squares: $x^2 + 2y^2 - 10x + 12y + 43 = (x - 5)^2 + 2(y + 3)^2 \geq 0$.
2. Therefore the expression is never negative.
3. Equality holds when $x = 5$, $y = -3$.

Final answer

$$x^2 + 2y^2 - 10x + 12y + 43 \geq 0$$

Worked Solutions

Inequalities

Q041 **Written****Prove that** $(a^2 + b^2)(x^2 + y^2) \geq (ax + by)^2$.**Solution**

1. Subtract the right-hand side: $(a^2 + b^2)(x^2 + y^2) - (ax + by)^2$.
2. Use $(a^2 + b^2)(x^2 + y^2) - (ax + by)^2 = (ay - bx)^2 \geq 0$.
3. Therefore the required inequality holds.
4. Equality holds when $ay = bx$.

Final answer

$$(a^2 + b^2)(x^2 + y^2) \geq (ax + by)^2$$

Worked Solutions

Inequalities

Q042 Challenge

For real x, y , prove that $x^2 + 2y^2 - 10x + 12y + 43 \geq 0$.

Solution

1. Complete both squares: $x^2 + 2y^2 - 10x + 12y + 43 = (x - 5)^2 + 2(y + 3)^2 \geq 0$.
2. Therefore $x^2 + 2y^2 - 10x + 12y + 43 \geq 0$.
3. Equality holds when $x = 5$, $y = -3$.

Final answer

0

Worked Solutions

Inequalities

Q043 Written

Prove that $\sqrt{\frac{a^2+b^2}{2}} \geq \frac{a+b}{2}$.

Solution

1. Both sides are positive. Compare their squares.

$$2. \left(\sqrt{\frac{a^2+b^2}{2}} \right)^2 - \left(\frac{a+b}{2} \right)^2 = \frac{(a-b)^2}{4} \geq 0.$$

3. Hence the required inequality holds.

4. Equality holds when $a = b$.

Final answer

$$\sqrt{\frac{a^2+b^2}{2}} \geq \frac{a+b}{2}$$

Worked Solutions

Inequalities

Q044 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + \frac{b^2}{2}} \geq \frac{a}{2} + \frac{b}{2}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Solution

1. Both sides are positive under the stated condition; compare their squares.
2. The difference of the squared sides is $\text{LHS}^2 - \text{RHS}^2 = \frac{(a-b)^2}{4}$.
3. This expression is non-negative under the stated conditions.
4. Therefore $\sqrt{\frac{a^2}{2} + \frac{b^2}{2}} \geq \frac{a}{2} + \frac{b}{2}$.
5. Equality occurs when the displayed square is zero.

Correct option: A

Final answer

$$\frac{(a-b)^2}{4}$$

Worked Solutions

Inequalities

Q045 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + 2b^2} \geq \frac{a}{2} + b$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Solution

1. Both sides are positive under the stated condition; compare their squares.
2. The difference of the squared sides is $\text{LHS}^2 - \text{RHS}^2 = \frac{(a-2b)^2}{4}$.
3. This expression is non-negative under the stated conditions.
4. Therefore $\sqrt{\frac{a^2}{2} + 2b^2} \geq \frac{a}{2} + b$.
5. Equality occurs when the displayed square is zero.

Correct option: A

Final answer

$$\frac{(a-2b)^2}{4}$$

Worked Solutions

Inequalities

Q046 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + \frac{9b^2}{2}} \geq \frac{a}{2} + \frac{3b}{2}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Solution

1. Both sides are positive under the stated condition; compare their squares.
2. The difference of the squared sides is $\text{LHS}^2 - \text{RHS}^2 = \frac{(a-3b)^2}{4}$.
3. This expression is non-negative under the stated conditions.
4. Therefore $\sqrt{\frac{a^2}{2} + \frac{9b^2}{2}} \geq \frac{a}{2} + \frac{3b}{2}$.
5. Equality occurs when the displayed square is zero.

Correct option: C

Final answer

$$\frac{(a-3b)^2}{4}$$

Worked Solutions

Inequalities

Q047 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + 8b^2} \geq \frac{a}{2} + 2b$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Solution

- Both sides are positive under the stated condition; compare their squares.
- The difference of the squared sides is $\text{LHS}^2 - \text{RHS}^2 = \frac{(a-4b)^2}{4}$.
- This expression is non-negative under the stated conditions.
- Therefore $\sqrt{\frac{a^2}{2} + 8b^2} \geq \frac{a}{2} + 2b$.
- Equality occurs when the displayed square is zero.

Correct option: A

Final answer

$$\frac{(a-4b)^2}{4}$$

Worked Solutions

Inequalities

Q048 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + \frac{25b^2}{2}} \geq \frac{a}{2} + \frac{5b}{2}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Solution

1. Both sides are positive under the stated condition; compare their squares.
2. The difference of the squared sides is $\text{LHS}^2 - \text{RHS}^2 = \frac{(a-5b)^2}{4}$.
3. This expression is non-negative under the stated conditions.
4. Therefore $\sqrt{\frac{a^2}{2} + \frac{25b^2}{2}} \geq \frac{a}{2} + \frac{5b}{2}$.
5. Equality occurs when the displayed square is zero.

Correct option: A

Final answer

$$\frac{(a-5b)^2}{4}$$

Worked Solutions

Inequalities

Q049 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + 18b^2} \geq \frac{a}{2} + 3b$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Solution

- Both sides are positive under the stated condition; compare their squares.
- The difference of the squared sides is $\text{LHS}^2 - \text{RHS}^2 = \frac{(a-6b)^2}{4}$.
- This expression is non-negative under the stated conditions.
- Therefore $\sqrt{\frac{a^2}{2} + 18b^2} \geq \frac{a}{2} + 3b$.
- Equality occurs when the displayed square is zero.

Correct option: B

Final answer

$$\frac{(a-6b)^2}{4}$$

Worked Solutions

Inequalities

Q050 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + \frac{49b^2}{2}} \geq \frac{a}{2} + \frac{7b}{2}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Solution

1. Both sides are positive under the stated condition; compare their squares.
2. The difference of the squared sides is $\text{LHS}^2 - \text{RHS}^2 = \frac{(a-7b)^2}{4}$.
3. This expression is non-negative under the stated conditions.
4. Therefore $\sqrt{\frac{a^2}{2} + \frac{49b^2}{2}} \geq \frac{a}{2} + \frac{7b}{2}$.
5. Equality occurs when the displayed square is zero.

Correct option: A

Final answer

$$\frac{(a-7b)^2}{4}$$

Worked Solutions

Inequalities

Q051 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + 32b^2} \geq \frac{a}{2} + 4b$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Solution

1. Both sides are positive under the stated condition; compare their squares.
2. The difference of the squared sides is $\text{LHS}^2 - \text{RHS}^2 = \frac{(a-8b)^2}{4}$.
3. This expression is non-negative under the stated conditions.
4. Therefore $\sqrt{\frac{a^2}{2} + 32b^2} \geq \frac{a}{2} + 4b$.
5. Equality occurs when the displayed square is zero.

Correct option: A

Final answer

$$\frac{(a-8b)^2}{4}$$

Worked Solutions

Inequalities

Q052 Written

Prove that $\sqrt{2(a+b)} \geq \sqrt{a} + \sqrt{b}$.**Solution**

1. Both sides are positive. Compare their squares.
2. $(\sqrt{2(a+b)})^2 - (\sqrt{a} + \sqrt{b})^2 = (\sqrt{a} - \sqrt{b})^2 \geq 0$.
3. Hence the required inequality holds.
4. Equality holds when $a = b$.

Final answer

$$\sqrt{2(a+b)} \geq \sqrt{a} + \sqrt{b}$$

Q053 Written

Prove that $a + 2 \geq \sqrt{3 + 6a}$.**Solution**

1. Both sides are positive. Compare their squares.
2. $(a + 2)^2 - (\sqrt{3 + 6a})^2 = (a - 1)^2 \geq 0$.
3. Hence the required inequality holds.
4. Equality holds when $a = 1$.

Final answer

$$a + 2 \geq \sqrt{3 + 6a}$$

Worked Solutions

Inequalities

Q054 Written

Prove that $\sqrt{2(a^2 + b^2)} \geq a + b$.

Solution

- Both sides are positive. Compare their squares.
- $(\sqrt{2(a^2 + b^2)})^2 - (a + b)^2 = (a - b)^2 \geq 0$.
- Hence the required inequality holds.
- Equality holds when $a = b$.

Final answer

$$\sqrt{2(a^2 + b^2)} \geq a + b$$

Q055 Written

Prove that $\sqrt{\frac{a+b}{2}} \geq \frac{\sqrt{a} + \sqrt{b}}{2}$.

Solution

- Both sides are positive. Compare their squares.
- $\left(\sqrt{\frac{a+b}{2}}\right)^2 - \left(\frac{\sqrt{a} + \sqrt{b}}{2}\right)^2 = \frac{(\sqrt{a} - \sqrt{b})^2}{4} \geq 0$.
- Hence the required inequality holds.
- Equality holds when $a = b$.

Final answer

$$\sqrt{\frac{a+b}{2}} \geq \frac{\sqrt{a} + \sqrt{b}}{2}$$

Worked Solutions

Inequalities

Q056 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{a} + \sqrt{b} > \sqrt{a + b}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Solution

1. Both sides are positive under the stated condition; compare their squares.
2. The difference of the squared sides is $\text{LHS}^2 - \text{RHS}^2 = 2\sqrt{a}\sqrt{b}$.
3. This expression is positive under the stated conditions.
4. Therefore $\sqrt{a} + \sqrt{b} > \sqrt{a + b}$.

Correct option: B

Final answer

$$2\sqrt{a}\sqrt{b}$$

Q057 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{a} + 2\sqrt{b} > \sqrt{a + 4b}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Solution

1. Both sides are positive under the stated condition; compare their squares.
2. The difference of the squared sides is $\text{LHS}^2 - \text{RHS}^2 = 4\sqrt{a}\sqrt{b}$.
3. This expression is positive under the stated conditions.
4. Therefore $\sqrt{a} + 2\sqrt{b} > \sqrt{a + 4b}$.

Correct option: A

Final answer

$$4\sqrt{a}\sqrt{b}$$

Worked Solutions

Inequalities

Q058 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{a} + 3\sqrt{b} > \sqrt{a + 9b}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Solution

1. Both sides are positive under the stated condition; compare their squares.
2. The difference of the squared sides is $\text{LHS}^2 - \text{RHS}^2 = 6\sqrt{a}\sqrt{b}$.
3. This expression is positive under the stated conditions.
4. Therefore $\sqrt{a} + 3\sqrt{b} > \sqrt{a + 9b}$.

Correct option: C

Final answer

$$6\sqrt{a}\sqrt{b}$$

Q059 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{a} + 4\sqrt{b} > \sqrt{a + 16b}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Solution

1. Both sides are positive under the stated condition; compare their squares.
2. The difference of the squared sides is $\text{LHS}^2 - \text{RHS}^2 = 8\sqrt{a}\sqrt{b}$.
3. This expression is positive under the stated conditions.
4. Therefore $\sqrt{a} + 4\sqrt{b} > \sqrt{a + 16b}$.

Correct option: B

Final answer

$$8\sqrt{a}\sqrt{b}$$

Worked Solutions

Inequalities

Q060 Written

Prove that $\sqrt{a} + 4\sqrt{b} > \sqrt{a + 16b}$.**Solution**

1. Both sides are positive. Compare their squares.
2. $(\sqrt{a} + 4\sqrt{b})^2 - (\sqrt{a + 16b})^2 = 8\sqrt{ab} > 0$.
3. Hence the strict inequality holds.
4. There is no equality case.

Final answer

$$\sqrt{a} + 4\sqrt{b} > \sqrt{a + 16b}$$

Q061 Written

Prove that $2\sqrt{a} + \sqrt{b} > \sqrt{4a + b}$.**Solution**

1. Both sides are positive. Compare their squares.
2. $(2\sqrt{a} + \sqrt{b})^2 - (\sqrt{4a + b})^2 = 4\sqrt{ab} > 0$.
3. Hence the strict inequality holds.
4. There is no equality case.

Final answer

$$2\sqrt{a} + \sqrt{b} > \sqrt{4a + b}$$

Worked Solutions

Inequalities

Q062 Written

Prove that $\sqrt{a} + \sqrt{b} > \sqrt{a+b}$.**Solution**

1. Both sides are positive. Compare their squares.
2. $(\sqrt{a} + \sqrt{b})^2 - (\sqrt{a+b})^2 = 2\sqrt{ab} > 0$.
3. Hence the strict inequality holds.
4. There is no equality case.

Final answer

$$\sqrt{a} + \sqrt{b} > \sqrt{a+b}$$

Q063 Written

Prove that $2\sqrt{a} + 3\sqrt{b} > \sqrt{4a+9b}$.**Solution**

1. Both sides are positive. Compare their squares.
2. $(2\sqrt{a} + 3\sqrt{b})^2 - (\sqrt{4a+9b})^2 = 12\sqrt{ab} > 0$.
3. Hence the strict inequality holds.
4. There is no equality case.

Final answer

$$2\sqrt{a} + 3\sqrt{b} > \sqrt{4a+9b}$$

Worked Solutions

Inequalities

Q064 **Written****Prove that** $\sqrt{a-b} > \sqrt{a} - \sqrt{b}$.**Solution**

1. Both sides are positive because $a > b > 0$. Compare their squares.
2. $(\sqrt{a-b})^2 - (\sqrt{a} - \sqrt{b})^2 = 2\sqrt{b}(\sqrt{a} - \sqrt{b}) > 0$.
3. Hence the strict inequality holds.
4. There is no equality case.

Final answer

$$\sqrt{a-b} > \sqrt{a} - \sqrt{b}$$

Q065 **Written****If** $a > b > 0$, **prove that** $\sqrt{a-b} < \sqrt{a} - \frac{b}{2\sqrt{a}}$.**Solution**

1. Both sides are positive. Compare their squares.
2. $\left(\sqrt{a} - \frac{b}{2\sqrt{a}}\right)^2 - (\sqrt{a-b})^2 = \frac{b^2}{4a} > 0$.
3. Therefore $\sqrt{a-b} < \sqrt{a} - \frac{b}{2\sqrt{a}}$.
4. There is no equality case.

Final answer

$$\sqrt{a-b} < \sqrt{a} - \frac{b}{2\sqrt{a}}$$

Worked Solutions

Inequalities

Q066 Challenge

For $a, b > 0$, prove that $\sqrt{a} + 4\sqrt{b} > \sqrt{a + 16b}$.

Solution

1. Both sides are positive under $a, b > 0$; compare their squares.
2. $(\sqrt{a} + 4\sqrt{b})^2 - (\sqrt{a + 16b})^2 = 8\sqrt{ab} > 0$.
3. Therefore $\sqrt{a} + 4\sqrt{b} > \sqrt{a + 16b}$.

Final answer

$$\sqrt{a + 16b}$$

Worked Solutions

Inequalities

Q067 Written

Prove that $\frac{b}{a} + \frac{a}{b} \geq 2$.**Solution**

1. By AM–GM, $\frac{b}{a} + \frac{a}{b} \geq 2\sqrt{\frac{b}{a} \cdot \frac{a}{b}} = 2$.
2. Equality holds when $a = b$.

Final answer

$$\frac{b}{a} + \frac{a}{b} \geq 2$$

Q068 Written

Prove that $\frac{b}{3a} + \frac{12a}{b} \geq 4$.**Solution**

1. Apply AM–GM to the two positive terms: $\frac{b}{3a} + \frac{12a}{b} \geq 2\sqrt{4} = 4$.
2. Equality holds when $b = 6a$.

Final answer

$$\frac{b}{3a} + \frac{12a}{b} \geq 4$$

Worked Solutions

Inequalities

Q069 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{a^2+b^2}{ab} \geq 2$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $\frac{a^2+b^2}{ab} - 2$.
2. Factor/complete the square to obtain $\frac{(a-b)^2}{ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{a^2+b^2}{ab} \geq 2$.

Correct option: A**Final answer**

$$\frac{(a-b)^2}{ab}$$

Q070 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{4a^2 + b^2}{2ab} \geq 2$ select the correct factorisation of LHS – RHS:

Solution

1. Start with LHS - RHS: $\frac{4a^2 + b^2}{2ab} - 2$.
2. Factor/complete the square to obtain $\frac{(2a - b)^2}{2ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{4a^2 + b^2}{2ab} \geq 2$.

Correct option: C

Final answer

$$\frac{(2a - b)^2}{2ab}$$

Worked Solutions

Inequalities

Q071 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{9a^2 + b^2}{3ab} \geq 2$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $\frac{9a^2 + b^2}{3ab} - 2$.
2. Factor/complete the square to obtain $\frac{(3a - b)^2}{3ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{9a^2 + b^2}{3ab} \geq 2$.

Correct option: B**Final answer**

$$\frac{(3a - b)^2}{3ab}$$

Q072 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{16a^2 + b^2}{4ab} \geq 2$ select the correct factorisation of LHS – RHS:

Solution

1. Start with LHS - RHS: $\frac{16a^2 + b^2}{4ab} - 2$.
2. Factor/complete the square to obtain $\frac{(4a - b)^2}{4ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{16a^2 + b^2}{4ab} \geq 2$.

Correct option: A

Final answer

$$\frac{(4a - b)^2}{4ab}$$

Worked Solutions

Inequalities

Q073 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{25a^2 + b^2}{5ab} \geq 2$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $\frac{25a^2 + b^2}{5ab} - 2$.
2. Factor/complete the square to obtain $\frac{(5a - b)^2}{5ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{25a^2 + b^2}{5ab} \geq 2$.

Correct option: D**Final answer**

$$\frac{(5a - b)^2}{5ab}$$

Q074 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{36a^2 + b^2}{6ab} \geq 2$ select the correct factorisation of LHS – RHS:

Solution

1. Start with LHS - RHS: $\frac{36a^2 + b^2}{6ab} - 2$.
2. Factor/complete the square to obtain $\frac{(6a - b)^2}{6ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{36a^2 + b^2}{6ab} \geq 2$.

Correct option: D

Final answer

$$\frac{(6a - b)^2}{6ab}$$

Worked Solutions

Inequalities

Q075 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{49a^2 + b^2}{7ab} \geq 2$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $\frac{49a^2 + b^2}{7ab} - 2$.
2. Factor/complete the square to obtain $\frac{(7a - b)^2}{7ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{49a^2 + b^2}{7ab} \geq 2$.

Correct option: C**Final answer**

$$\frac{(7a - b)^2}{7ab}$$

Q076 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{64a^2 + b^2}{8ab} \geq 2$ select the correct factorisation of LHS – RHS:

Solution

1. Start with LHS - RHS: $\frac{64a^2 + b^2}{8ab} - 2$.
2. Factor/complete the square to obtain $\frac{(8a - b)^2}{8ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{64a^2 + b^2}{8ab} \geq 2$.

Correct option: D

Final answer

$$\frac{(8a - b)^2}{8ab}$$

Worked Solutions

Inequalities

Q077 Written

Prove that $(a + b)\left(\frac{1}{a} + \frac{1}{b}\right) \geq 4$.**Solution**

1. Expand: $(a + b)\left(\frac{1}{a} + \frac{1}{b}\right) = \frac{a}{b} + \frac{b}{a} + 2$.
2. Use $\frac{a}{b} + \frac{b}{a} \geq 2$ to obtain the lower bound 4.
3. Equality holds when $a = b$.

Final answer

$$(a + b)\left(\frac{1}{a} + \frac{1}{b}\right) \geq 4$$

Q078 Written

Prove that $\left(x + \frac{1}{x}\right)\left(x + \frac{4}{x}\right) \geq 9$.**Solution**

1. Expand to $x^2 + 5 + \frac{4}{x^2}$.
2. Since $x^2 + \frac{4}{x^2} \geq 4$, the product is at least 9.
3. Equality holds when $x = \sqrt{2}$.

Final answer

$$\left(x + \frac{1}{x}\right)\left(x + \frac{4}{x}\right) \geq 9$$

Worked Solutions

Inequalities

Q079 Written

Prove that $(a + 8b)\left(\frac{2}{a} + \frac{1}{b}\right) \geq 18$.**Solution**

1. Expand to $10 + \frac{a}{b} + \frac{16b}{a}$.
2. AM–GM gives $\frac{a}{b} + \frac{16b}{a} \geq 8$, so the product is at least 18.
3. Equality holds when $a = 4b$.

Final answer

$$(a + 8b)\left(\frac{2}{a} + \frac{1}{b}\right) \geq 18$$

Q080 Written

Prove that $\left(a + \frac{3}{b}\right)\left(b + \frac{3}{a}\right) \geq 12$.**Solution**

1. Expand to $ab + 6 + \frac{9}{ab}$.
2. AM–GM gives $ab + \frac{9}{ab} \geq 6$, so the product is at least 12.
3. Equality holds when $ab = 3$.

Final answer

$$\left(a + \frac{3}{b}\right)\left(b + \frac{3}{a}\right) \geq 12$$

Worked Solutions

Inequalities

Q081 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{(a+b)^2}{ab} \geq 4$ select the correct factorisation of LHS – RHS:

Solution

1. Start with LHS - RHS: $\frac{(a+b)^2}{ab} - 4$.
2. Factor/complete the square to obtain $\frac{(a-b)^2}{ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{(a+b)^2}{ab} \geq 4$.

Correct option: D**Final answer**

$$\frac{(a-b)^2}{ab}$$

Q082 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{(a+4b)(4a+b)}{ab} \geq 25$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $\frac{(a+4b)(4a+b)}{ab} - 25$.
2. Factor/complete the square to obtain $\frac{4(a-b)^2}{ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{(a+4b)(4a+b)}{ab} \geq 25$.

Correct option: B

Final answer

$$\frac{4(a-b)^2}{ab}$$

Worked Solutions

Inequalities

Q083 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{(a+9b)(9a+b)}{ab} \geq 100$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $\frac{(a+9b)(9a+b)}{ab} - 100$.
2. Factor/complete the square to obtain $\frac{9(a-b)^2}{ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{(a+9b)(9a+b)}{ab} \geq 100$.

Correct option: C**Final answer**

$$\frac{9(a-b)^2}{ab}$$

Q084 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{(a+16b)(16a+b)}{ab} \geq 289$ select the correct factorisation of LHS – RHS:

Solution

1. Start with LHS - RHS: $\frac{(a+16b)(16a+b)}{ab} - 289$.
2. Factor/complete the square to obtain $\frac{16(a-b)^2}{ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{(a+16b)(16a+b)}{ab} \geq 289$.

Correct option: D

Final answer

$$\frac{16(a-b)^2}{ab}$$

Worked Solutions

Inequalities

Q085 Written

Prove that $\left(a + \frac{1}{b}\right)\left(b + \frac{9}{a}\right) \geq 16$.**Solution**

1. Expand to $ab + 10 + \frac{9}{ab}$.
2. AM–GM gives $ab + \frac{9}{ab} \geq 6$, so the product is at least 16.
3. Equality holds when $ab = 3$.

Final answer

$$\left(a + \frac{1}{b}\right)\left(b + \frac{9}{a}\right) \geq 16$$

Q086 Written

Prove that $a + \frac{4}{a} \geq 4$.**Solution**

1. Apply AM–GM: $a + \frac{4}{a} \geq 2\sqrt{4} = 4$.
2. Equality holds when $a = 2$.

Final answer

$$a + \frac{4}{a} \geq 4$$

Worked Solutions

Inequalities

Q087 Written

Prove that $3ab + \frac{4}{ab} \geq 4\sqrt{3}$.**Solution**

1. Apply AM–GM to $3ab$ and $\frac{4}{ab}$: the sum is at least $2\sqrt{12} = 4\sqrt{3}$.
2. Equality holds when $ab = \frac{2}{\sqrt{3}}$.

Final answer

$$3ab + \frac{4}{ab} \geq 4\sqrt{3}$$

Q088 Written

Prove that $a + b + \frac{9}{a+b} \geq 6$.**Solution**

1. Let $t = a + b > 0$. AM–GM gives $t + \frac{9}{t} \geq 6$.
2. Equality holds when $a + b = 3$.

Final answer

$$a + b + \frac{9}{a+b} \geq 6$$

Worked Solutions

Inequalities

Q089 Written

A rectangle has perimeter 200 cm. Prove that its area $LW \leq 2500 \text{ cm}^2$.

Solution

1. From the perimeter, $L + W = 100$.
2. AM–GM gives $\frac{L+W}{2} \geq \sqrt{LW}$, so $50 \geq \sqrt{LW}$.
3. Therefore $LW \leq 2500$.
4. Equality holds when $L = W = 50$.

Final answer

$$LW \leq 2500$$

Worked Solutions

Inequalities

Q090 Challenge

For $a, b > 0$, prove that $\frac{b}{a} + \frac{a}{b} \geq 2$.

Solution

1. Put the terms over a positive denominator: $\frac{b}{a} + \frac{a}{b} - 2 = \frac{(a-b)^2}{ab} \geq 0$.
2. Hence $\frac{b}{a} + \frac{a}{b} \geq 2$.
3. Equality holds when $a = b$.

Final answer

2

Worked Solutions

Inequalities

Quiz MCQ 1 Concept Quiz · MCQ

If $x > y$, prove that $3x + y > 2(x + y)$.

If $x > y$, the positive difference between the two sides is:

Solution

1. Subtract the right-hand side: $x - y$.
2. Factor the difference as $x - y$.
3. The condition $x > y$ makes each factor positive.
4. Hence $3x + y > 2(x + y)$.

Correct option: A

Final answer

$$x - y$$

Quiz MCQ 2 Concept Quiz · MCQ

Under a, b real, prove that $a^2 + 25b^2 \geq 10ab$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $a^2 + 25b^2 - 10ab$.
2. Factor/complete the square to obtain $(a - 5b)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $a^2 + 25b^2 \geq 10ab$.

Correct option: A

Final answer

$$(a - 5b)^2$$

Worked Solutions

Inequalities

Quiz MCQ 3 Concept Quiz · MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + \frac{49b^2}{2}} \geq \frac{a}{2} + \frac{7b}{2}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Solution

1. Both sides are positive under the stated condition; compare their squares.
2. The difference of the squared sides is $\text{LHS}^2 - \text{RHS}^2 = \frac{(a-7b)^2}{4}$.
3. This expression is non-negative under the stated conditions.
4. Therefore $\sqrt{\frac{a^2}{2} + \frac{49b^2}{2}} \geq \frac{a}{2} + \frac{7b}{2}$.
5. Equality occurs when the displayed square is zero.

Correct option: A

Final answer

$$\frac{(a-7b)^2}{4}$$

Worked Solutions

Inequalities

Quiz MCQ 4 Concept Quiz · MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{(a+4b)(4a+b)}{ab} \geq 25$ select the correct factorisation of LHS – RHS:

Solution

1. Start with LHS - RHS: $\frac{(a+4b)(4a+b)}{ab} - 25$.
2. Factor/complete the square to obtain $\frac{4(a-b)^2}{ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{(a+4b)(4a+b)}{ab} \geq 25$.

Correct option: B

Final answer

$$\frac{4(a-b)^2}{ab}$$

Quiz WRITTEN 5 Concept Quiz · Written

If $x > y$, prove that $3x + 4y > 2x + 5y$.

Solution

1. Subtract the right-hand side: $x - y$.
2. Factor the difference as $x - y$.
3. The condition $x > y$ makes each factor positive.
4. Hence $3x + 4y > 2x + 5y$.

Final answer

$$2x + 5y$$

Worked Solutions

Inequalities

Quiz WRITTEN 6 Concept Quiz · Written

Under x, y real, prove that $x^2 + y^2 \geq 2xy$.

Solution

1. Start with LHS - RHS: $x^2 + y^2 - 2xy$.
2. Factor/complete the square to obtain $(x - y)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $x^2 + y^2 \geq 2xy$.

Final answer

$$2xy$$

Quiz WRITTEN 7 Concept Quiz · Written

Under $a > 0$ and $b > 0$, prove that $\sqrt{a} + 6\sqrt{b} > \sqrt{a + 36b}$.

Solution

1. Both sides are positive under the stated condition; compare their squares.
2. The difference of the squared sides is $\text{LHS}^2 - \text{RHS}^2 = 12\sqrt{a}\sqrt{b}$.
3. This expression is positive under the stated conditions.
4. Therefore $\sqrt{a} + 6\sqrt{b} > \sqrt{a + 36b}$.

Final answer

$$\sqrt{a + 36b}$$

Worked Solutions

Inequalities

Quiz WRITTEN 8 Concept Quiz · Written

Under $t > 0$, prove that $\frac{t^2+1}{t} \geq 2$.

Solution

1. Start with LHS - RHS: $\frac{t^2+1}{t} - 2$.
2. Factor/complete the square to obtain $\frac{(t-1)^2}{t}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{t^2+1}{t} \geq 2$.

Final answer

2

Answer key

Use the question number shown on each page.

Q001 $2(ax + by) > (a + b)(x + y)$

Q002 A

Q003 C

Q004 D

Q005 D

Q006 B

Q007 $ac + bd > ad + bc$

Q008 $ab + cd > ad + bc$

Q009 D

Q010 C

Q011 A

Q012 $3x - 4y > 2x - 3y$

Q013 $xy + 5x > 3y + 15$

Q014 $xy + 2 > x + 2y$

Q015 $ab + 1 > a + b$

Q016 $ax + by > ay + bx$

Q017 $ax + by + cz > ay + bz + cx$

Q018 D

Q019 $a^2 + 4b^2 \geq 4ab$

Q020 D

Q021 C

Q022 B

Q023 A

Q024 C

Q025 B

Q026 $a^2 + 5b^2 \geq 4ab$

Q027 $9x^2 \geq y(6x - y)$

Q028 $2(x^2 + y^2) \geq (x + y)^2$

Q029 $(2x - y)^2 \geq 3(x^2 - y^2)$

Q030 $(2a + 3b)^2 \geq 12ab$

Q031 A

Q032 C

Q033 D

Q034 D

Q035 A

Q036 C

Q037 $x^2 + y^2 \geq 2(x - y - 1)$

Q038 $x^2 + 2 > 2x$

Q039 $x^2 + y^2 - 6x + 8y + 25 \geq 0$

Q040 $x^2 + 2y^2 - 10x + 12y + 43 \geq 0$

Q041 $(a^2 + b^2)(x^2 + y^2) \geq (ax + by)^2$

Q042 0

Q043 $\sqrt{\frac{a^2 + b^2}{2}} \geq \frac{a + b}{2}$

Q044 A

Q045 A

Q046 C

Q047 A

Q048 A

Q049 B

Q050 A

Q051 A

Q052 $\sqrt{2(a + b)} \geq \sqrt{a} + \sqrt{b}$

Q053 $a + 2 \geq \sqrt{3 + 6a}$

Q054 $\sqrt{2(a^2 + b^2)} \geq a + b$

Q055 $\sqrt{\frac{a + b}{2}} \geq \frac{\sqrt{a} + \sqrt{b}}{2}$

Q056 B

Q057 A

Q058 C

Q059 B

Q060 $\sqrt{a} + 4\sqrt{b} > \sqrt{a + 16b}$

Q061 $2\sqrt{a} + \sqrt{b} > \sqrt{4a + b}$

Q062 $\sqrt{a} + \sqrt{b} > \sqrt{a + b}$

Q063 $2\sqrt{a} + 3\sqrt{b} > \sqrt{4a + 9b}$

Q064 $\sqrt{a - b} > \sqrt{a} - \sqrt{b}$

Q065 $\sqrt{a - b} < \sqrt{a} - \frac{b}{2\sqrt{a}}$

Q066 $\sqrt{a + 16b}$

Q067 $\frac{b}{a} + \frac{a}{b} \geq 2$

Q068 $\frac{b}{3a} + \frac{12a}{b} \geq 4$

Q069 A

Q070 C

Q071 B

Q072 A

Q073 D

Q074 D

Q075 C

Q076 D

Q077 $(a + b)\left(\frac{1}{a} + \frac{1}{b}\right) \geq 4$

Q078 $\left(x + \frac{1}{x}\right)\left(x + \frac{4}{x}\right) \geq 9$

Q079 $(a + 8b)\left(\frac{2}{a} + \frac{1}{b}\right) \geq 18$

Q080 $\left(a + \frac{3}{b}\right)\left(b + \frac{3}{a}\right) \geq 12$

Q081 D

Q082 B

Q083 C

Q084 D

Q085 $\left(a + \frac{1}{b}\right)\left(b + \frac{9}{a}\right) \geq 16$

Q086 $a + \frac{4}{a} \geq 4$

Q087 $3ab + \frac{4}{ab} \geq 4\sqrt{3}$

Q088 $a + b + \frac{9}{a + b} \geq 6$

Q089 $LW \leq 2500$

Q090 2

Quiz MCQ 1 A

Quiz MCQ 2 A

Quiz MCQ 3 A

Quiz MCQ 4 B

Quiz WRITTEN 5 $2x + 5y$

Quiz WRITTEN 6 $2xy$

Quiz WRITTEN 7 $\sqrt{a + 36b}$

Quiz WRITTEN 8 2